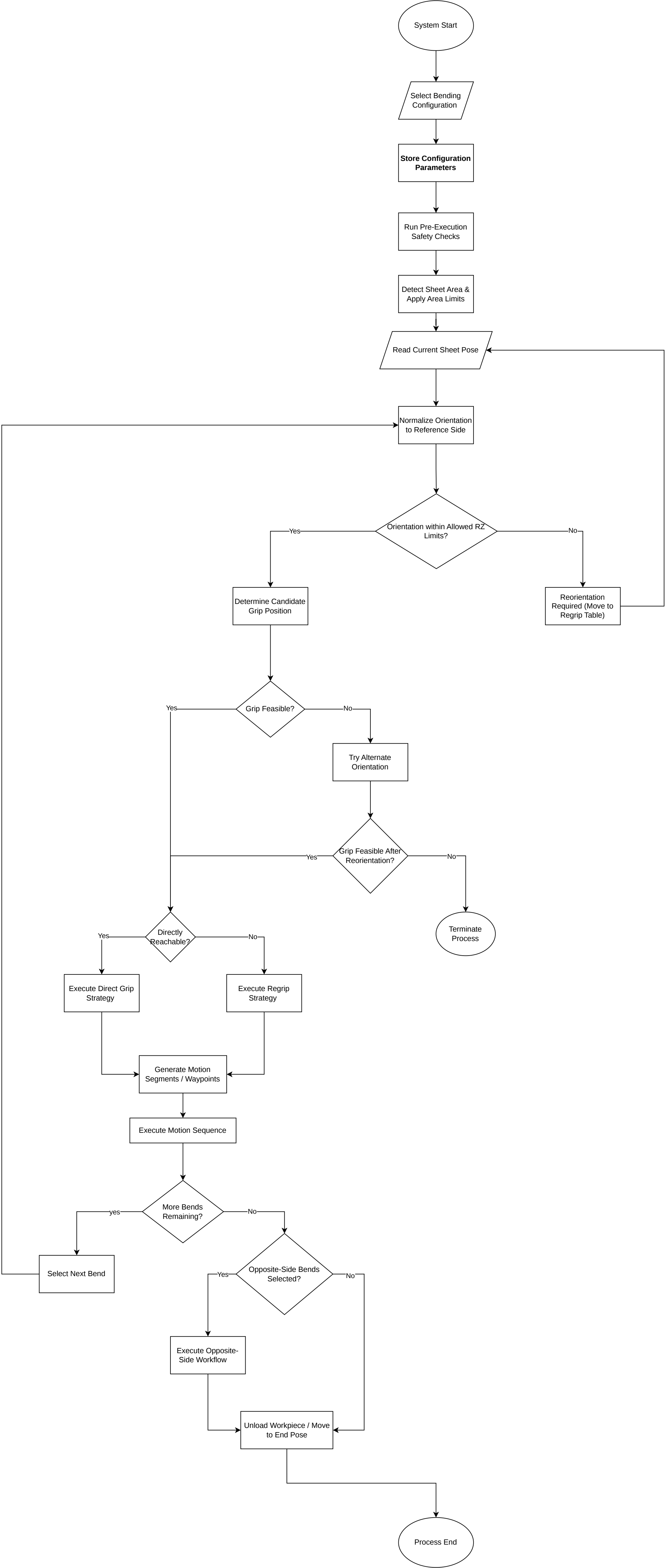
System Start
Select Bending Configuration
Store Configuration Parameters
Run Pre-Execution Safety Checks
Detect Sheet Area & Apply Area Limits
Read Current Sheet Pose
Normalize Orientation to Reference Side
Orientation within Allowed RZ Limits?
Yes
No
Determine Candidate Grip Position
Reorientation Required (Move to Regrip Table)
Grip Feasible?
Yes
No
Try Alternate Orientation
Grip Feasible After Reorientation?
Yes
No
Terminate Process
Directly Reachable?
Yes
No
Execute Direct Grip Strategy
Execute Regrip Strategy
Generate Motion Segments / Waypoints
Execute Motion Sequence
More Bends Remaining?
yes
No
Select Next Bend
Opposite-Side Bends Selected?
Yes
No
Execute Opposite-Side Workflow
Unload Workpiece / Move to End Pose
Process End